

Distributivity and the Commensurability of Value-Definite Realism and Empirical Adequacy

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Abstract

When does an empirically adequate theory admit a value-definite completion — an assignment of simultaneous definite values to all observables? We show that the answer depends on the algebraic structure of the observations. For a directed system of Boolean algebras carrying compatible charges, the Stone construction produces an unconditional measure on the dual space, and every observable admits a simultaneous definite value via ultrafilters. The passage to a probability measure on an underlying state space requires only σ -additivity, and the choice of state space is unconstrained by the algebra [4].

For orthomodular lattices — the algebraic framework for incompatible observations — the situation is structurally different. The McDonald–Bimbó dual space carries principal filters as distinguished structure, and the Kochen–Specker theorem blocks global value-definiteness. Probabilistic realism (descent to a probability measure) remains available via Gleason-type results, but value-definite realism does not.

The dividing line is distributivity. We identify a four-level coherence hierarchy — consistency, contextual coherence, probabilistic coherence, value-definite coherence — and show that distributivity of the observation algebra is the exact condition under which all four levels are mutually commensurable.

2020 Mathematics Subject Classification. 03G12 (primary); 06C15, 81P10, 60A10 (secondary).

Keywords. orthomodular lattice, Stone duality, value-definiteness, Kochen–Specker theorem, Gleason’s theorem, empirical adequacy, constructive empiricism, Boolean algebra, directed system, σ -additivity.

1 Introduction

A central question in the philosophy of science asks when an empirically successful theory can be completed to a realist one — one that assigns definite properties to an underlying reality, not merely adequate predictions to observable phenomena. Van Fraassen [5] argues that science need only aim at *empirical adequacy*: agreement with all observable phenomena. The scientific realist demands more — an underlying state space whose properties generate the observations.

This debate is usually conducted in philosophical terms. We give it a mathematical formulation and show that its resolution depends on the algebraic structure of the observations.

Three kinds of realism

We distinguish three nested positions, each a strengthening of the previous:

- (i) **Empirical adequacy (EA).** A measure on the dual space of the observation algebra that agrees with all observed charges. No commitment to an underlying state space.
- (ii) **Probabilistic realism (PR).** A probability measure on an underlying state space $(\Omega, \sigma(\mathcal{C}))$ that generates the observations. Requires descent from the dual-space measure.

- (iii) **Value-definite realism (VDR).** Every observable simultaneously has a definite value. A global dispersion-free state — equivalently, a 2-valued homomorphism on the full observation algebra.

These are nested: VDR implies PR implies EA. The question is which implications reverse, and what governs the answer.

The role of the observation algebra

Consider two settings:

Compatible observations. When all observations can be performed simultaneously, the observation algebra is a Boolean algebra — a complemented distributive lattice. Every Boolean algebra admits 2-valued homomorphisms (ultrafilters exist by Zorn’s lemma). In this setting, all three positions are available and mutually commensurable. The passage from EA to PR requires σ -additivity of the charges [4]; the further passage to VDR is free.

Incompatible observations. When some observations cannot be performed simultaneously — as in quantum mechanics, contextual experimental design, or distributed systems with interfering sensors — the observation algebra is an orthomodular lattice (OML), a complemented but non-distributive lattice. The Kochen–Specker theorem [3] shows that for the prototypical case $L(H)$ with $\dim H \geq 3$, no 2-valued homomorphism exists: VDR is blocked. Probabilistic realism remains available via Gleason’s theorem [2], and empirical adequacy is unconditional.

The dividing line is distributivity.

Contributions

We develop the following:

1. A unified framework based on directed systems of complemented lattices, with the Boolean and orthomodular cases as instances (§2–§??).
2. A four-level coherence hierarchy — consistency, contextual coherence, probabilistic coherence, value-definite coherence — with the obstruction at each level identified (§??).
3. A structural theorem: distributivity of the observation algebra is the exact condition under which all four levels are mutually commensurable (§??).
4. An analysis of the philosophical consequences: the realism/empiricism debate has different mathematical answers depending on the observation algebra (§??).

2 The Boolean case

We recall the framework and results of [4], which treats directed systems of Boolean algebras.

Directed systems and compatible charges

Let $(\iota, \leq)$ be a directed set indexing a family of Boolean algebras $\{B_i\}_{i \in \iota}$ with injective homomorphisms $\varphi_{ij} : B_i \rightarrow B_j$ for $i \leq j$. Write $\mathcal{C} = \bigcup_i B_i$ for the direct limit — the *cylinder algebra*.

A *compatible family of normalised charges* is a collection $\{\ell_i\}$ with $\ell_i : B_i \rightarrow [0, 1]$ finitely additive, $\ell_i(1) = 1$, and $\ell_j \circ \varphi_{ij} = \ell_i$ for $i \leq j$.

The Stone construction

The Stone space $\text{St}(\mathcal{C})$ is the space of ultrafilters on $\mathcal{C}$, compact and totally disconnected. Each $\omega \in \Omega$ determines a principal ultrafilter $\text{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$.

Any compatible family of normalised charges extends to a σ -additive Baire probability measure $\hat{\mu}$ on $\text{St}(\mathcal{C})$ — unconditionally, with no assumption of σ -additivity [1, 4]. This is the *empirically adequate* object: it agrees with all charges on cylinder events.

Descent and σ -additivity

The measure $\hat{\mu}$ lives on $\text{St}(\mathcal{C})$, not on Ω . Descent to a probability measure P on $(\Omega, \sigma(\mathcal{C}))$ requires $\hat{\mu}(\text{pure}(\Omega)) = 1$ — that $\hat{\mu}$ concentrates on the principal ultrafilters. This holds if and only if each charge ℓ_i is σ -additive [4].

Value-definiteness in the Boolean case

Every Boolean algebra admits ultrafilters (by Zorn's lemma), and every ultrafilter is a 2-valued homomorphism. A 2-valued homomorphism assigns to each element $a \in \mathcal{C}$ a definite value in $\{0, 1\}$, respecting the lattice operations.

Moreover, any subset $S \subseteq \text{St}(\mathcal{C})$ that projects surjectively onto each $\text{St}(B_i)$ can serve as a realisation space: set $\Omega = S$ with evaluation maps given by the restriction maps. The algebra imposes no constraint on the choice of Ω [4].

Proposition 2.1 (Boolean commensurability). *For a directed system of Boolean algebras with compatible charges:*

- (a) *EA is unconditional: $\hat{\mu}$ on $\text{St}(\mathcal{C})$ always exists.*
- (b) *PR holds iff each ℓ_i is σ -additive.*
- (c) *VDR holds unconditionally: $\text{St}(\mathcal{C})$ consists entirely of 2-valued homomorphisms.*

In particular, EA, PR, and VDR are mutually commensurable: the only obstruction is σ -additivity at the EA–PR transition, and it is a condition on the charges, not the algebra.

Proof. Part (a) is the Stone construction [1, 4]. Part (b) is [4, Theorem 1]. Part (c): ultrafilters on a Boolean algebra are exactly the 2-valued homomorphisms; their existence follows from Zorn's lemma applied to the filter generated by any non-zero element. $\square$

References

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